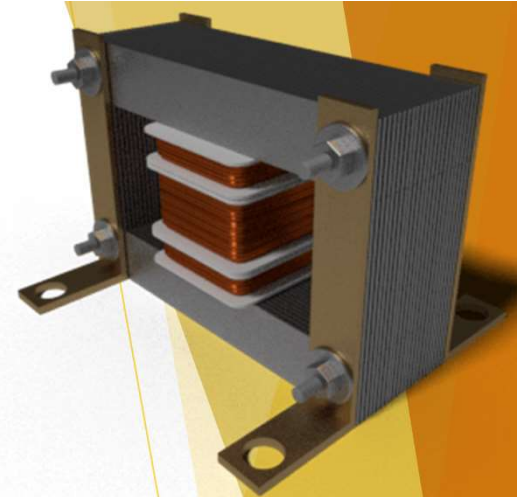




Transformer

Md. Kamrul Hasan

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Reference Book

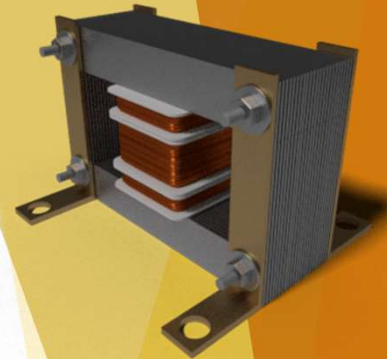
- ▶ 1. Electric Machinery Fundamentals- **Stephen J. Chapman**
- ▶ 2. Principles of Electrical Machines- **V.K Mehta**
- ▶ 3. A text book of Electrical Technology- **B.L. Theraja**

ATTENTION!!!!!!

This presentation slide only contains the overview of the related topics. Students are advised to take decent class-notes and read thoroughly from the prescribed text books.

Outline

- ▶ Introduction
- ▶ Why important ?
- ▶ Construction
- ▶ Working Principle
- ▶ Losses in transformer
- ▶ Equivalent Circuit
- ▶ Short Circuit Test
- ▶ Open Circuit Test
- ▶ Three Phase Transformer
- ▶ References

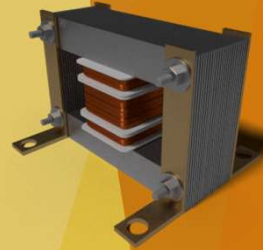


Transformer



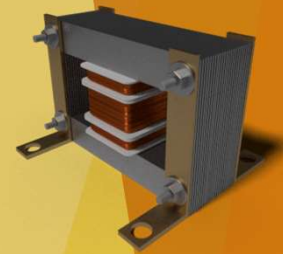
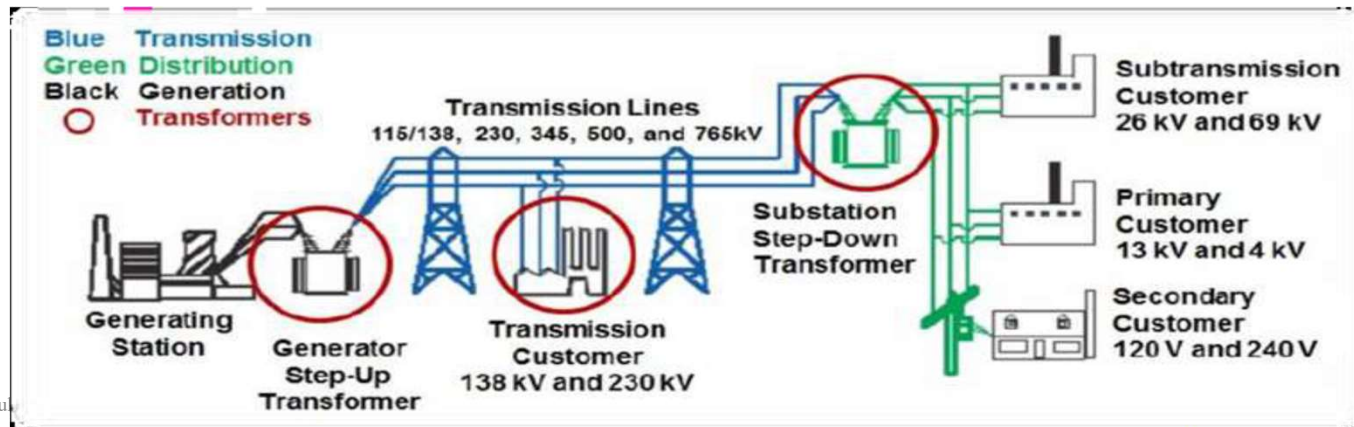
Md. Kamrul Hasan, EEE, CUET

- ▶ Transformer is an electrical devices that changes the **magnitude of alternating voltage or current** from one value to another.
- ▶ It has **no moving parts** so it is a static device.
- ▶ Because of very **rugged and durable in construction** it requires very little maintenance.



Why Important ?

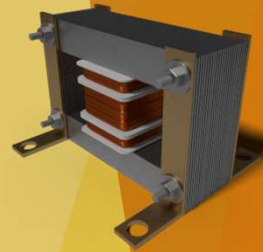
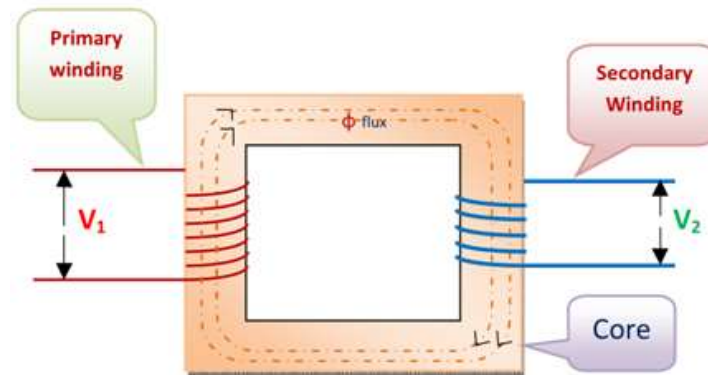
- ▶ Transformers are used to **adjust the voltages** to a proper level.
- ▶ It **transfer electrical energy** without changing the **frequency**.
- ▶ The transformers are the basic components for the **transmission of the electricity**.
- ▶ By increasing the voltages, the loss of the electricity in the transmission line is reduced.
- ▶ Transformer is used to increase the voltage at the **power generating station** (Step up) and used to decrease the voltage (Step down) for **house hold purpose**.



Transformer Construction

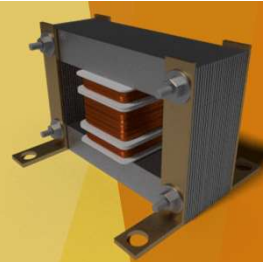
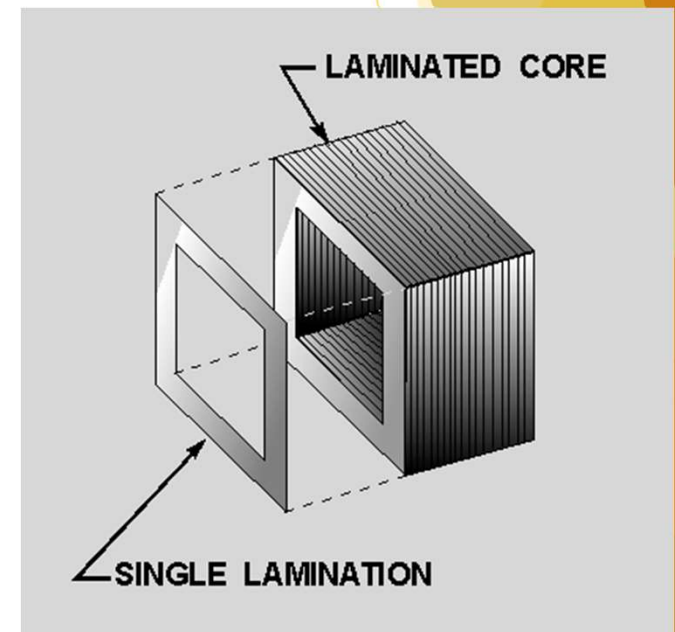
► Transformers have mainly three parts—

1. Core
2. Primary winding and
3. Secondary winding



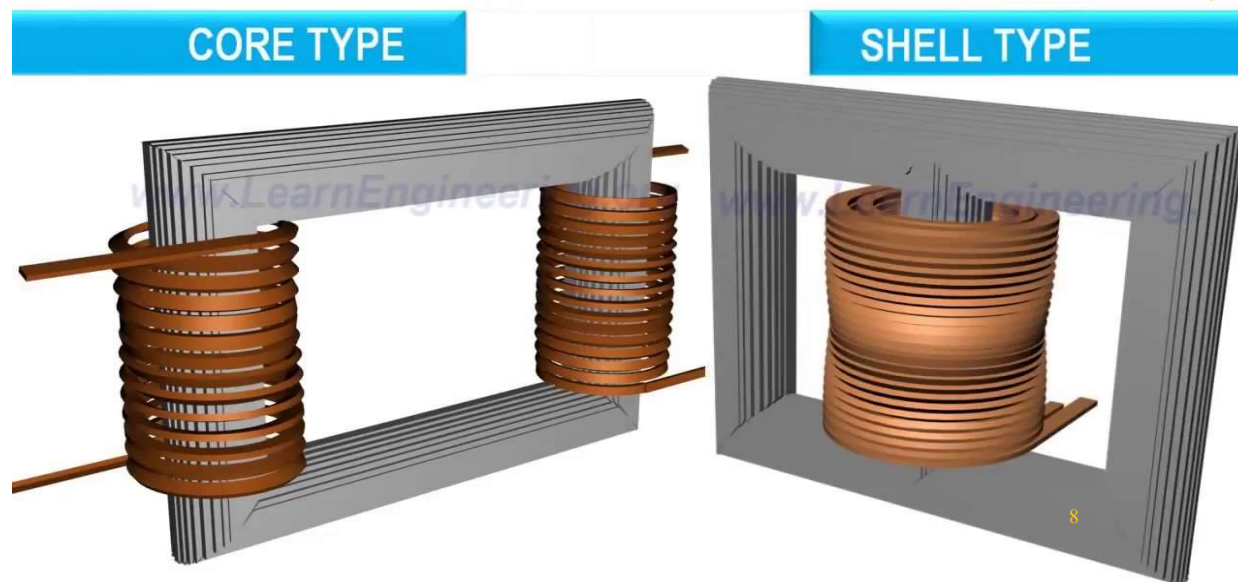
Transformer Core

- ▶ The main problem with the **transformer core** is its ***hysteresis and eddy current*** losses. To reduce the losses the core is construction with laminated steel sheet which is
 - **high silicon content,**
 - **high permeability, and**
 - **low hysteresis loss.**
- ▶ For minimizing **the eddy current loss**, the steel sheet is laminated each other by a light coat of core-plate varnish.



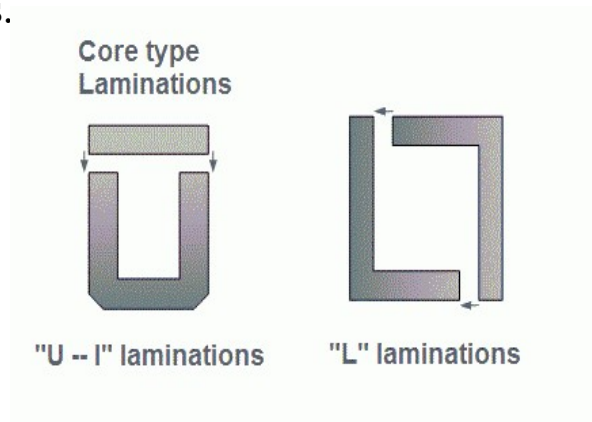
Transformer Core

- ▶ The transformer are two types according to core-
 1. Core type.
 2. Shell type.



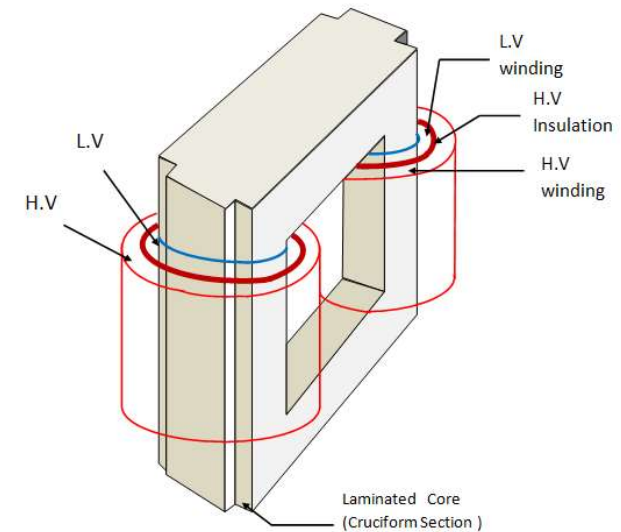
Core Type Transformer

- ▶ In Core type transformer, the windings surround a considerable part of the core.
- ▶ In the core type transformer magnetic core is made from 'L' shape strips.

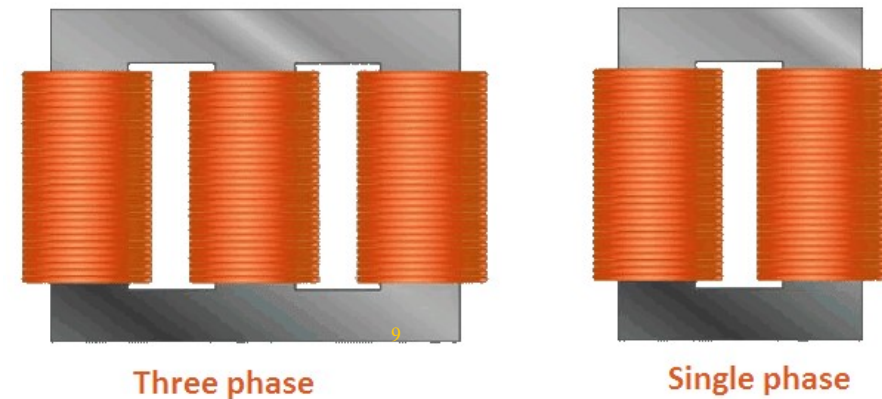


- ▶ For reducing leakage flux high voltage and low voltage winding are placed one after another separating with high insulation.

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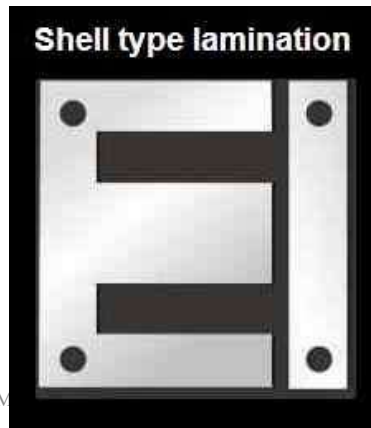


Core Type Transformer

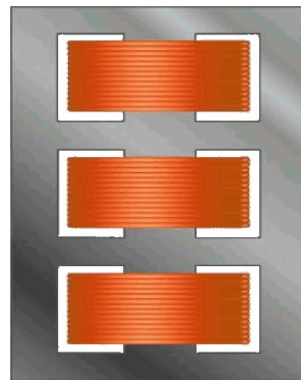


Shell type Transformer

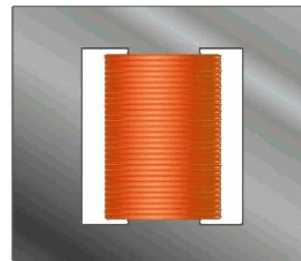
- ▶ In Shell type transformer, the core surround a considerable part of the winding.
- ▶ The core of shell type transformer is made from 'E' and 'I' strips.
- ▶ The width of the central limb is more than that of the outer limbs carries the whole of the flux.



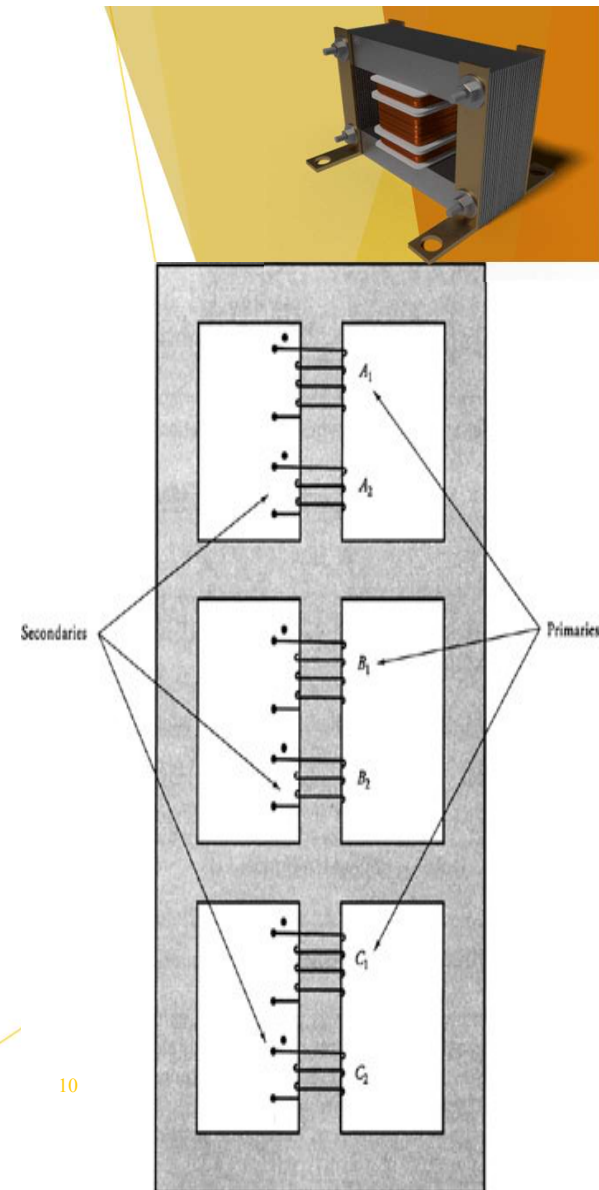
Shell Type Transformer



Three phase

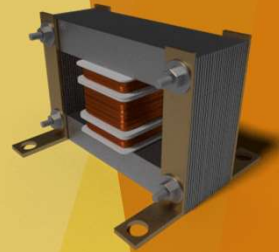
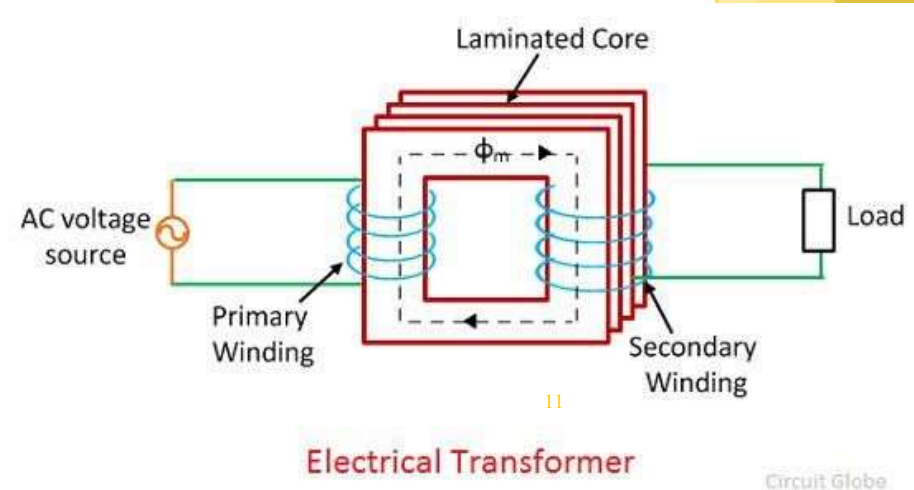
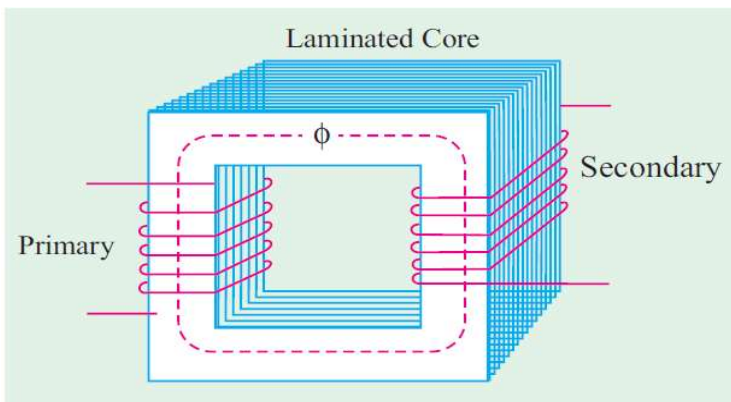


Single phase



Working Principle of Transformer

- ▶ It essentially consists of *two windings*.
- ▶ The primary and secondary winding wound on a common laminated magnetic core.
- ▶ The winding connected to the ac source is called primary winding.
- ▶ The one connected to load is called secondary winding (or secondary).



Working Principle of Transformer

Transformer works on the principle of **Faraday's Electromagnetic Mutual Induction**.

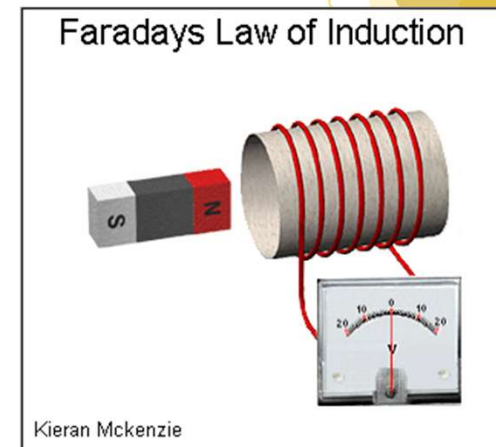
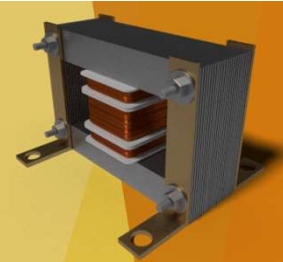
Faraday's First Law

"Whenever a conductor is placed in a varying magnetic field, an electromotive force (emf)/(voltage) is induced. If the conductor circuit is closed, a current is induced which is called induced current."

Faraday's Second Law

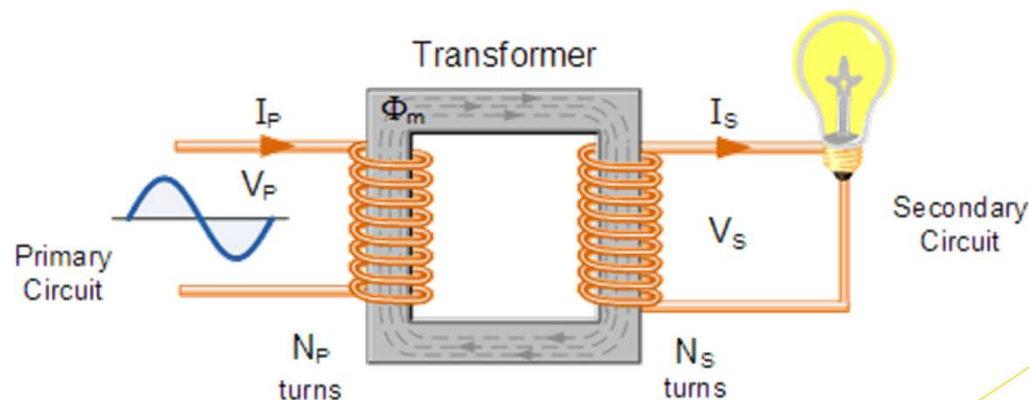
"The induced e.m.f. in a coil is equal to the rate of change of flux linkage."

$$E = N \frac{d\Phi}{dt}$$



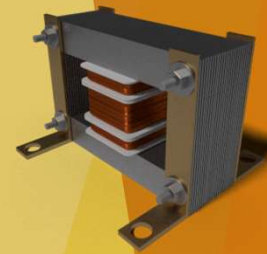
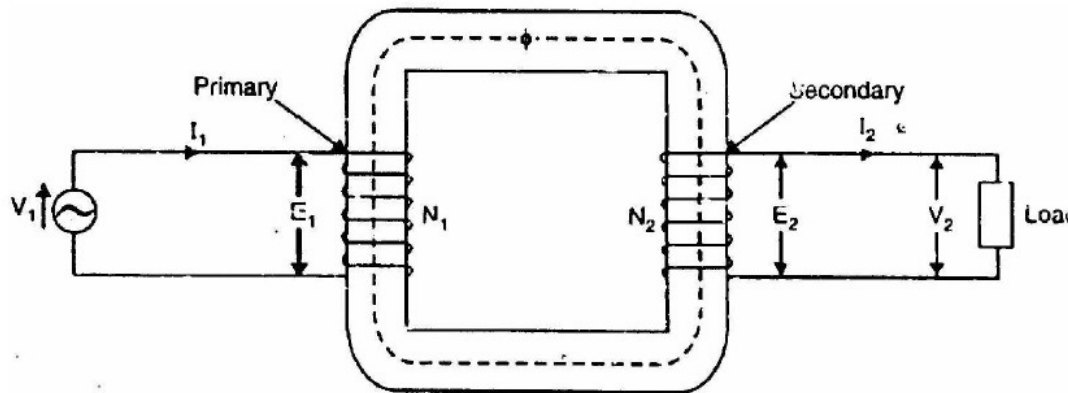
Working Principle of Transformer

- ▶ The basic of a transformer is **mutual induction** between two circuits linked by a common magnetic flux.
- ▶ If one coil of transformer is connected to a source of alternating voltage, an alternating flux is set up in the laminated core, most of the flux is linked with the other coil, according to Faraday's law of electromagnetic induction.



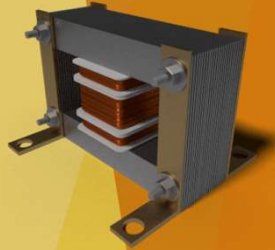
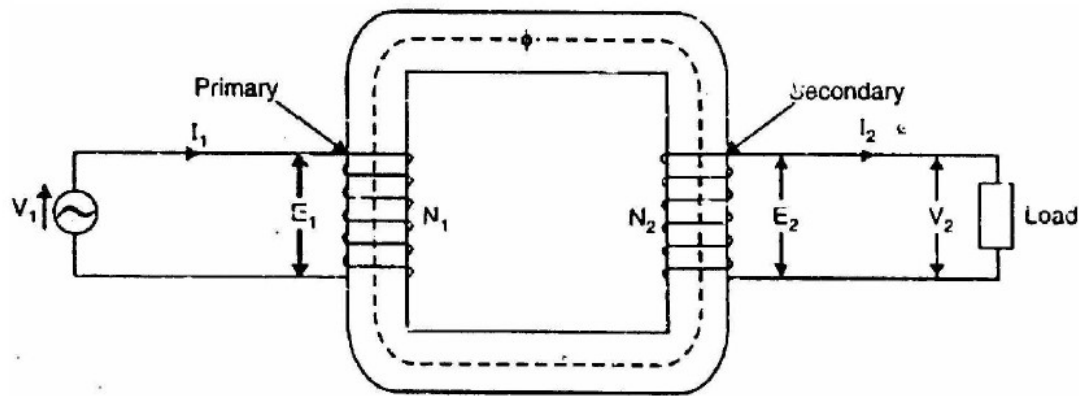
Working Principle of Transformer

- ▶ The alternating **voltage** V_1 is applied to the primary.
- ▶ Depending upon the number of turns of the primary (N_1) an alternating flux ϕ is set up in the core.
- ▶ This alternating flux ϕ links both the windings and induces e.m.f.s E_1 and E_2 in them according to Faraday's laws of electromagnetic induction.



Working Principle of Transformer

- ▶ If load is connected across the secondary winding, the secondary **e.m.f. E_2** will cause a **current I_2** to flow through the load.
- ▶ Consequently, **terminal voltage V_2** will appear across the load.



Working Principle of Transformer

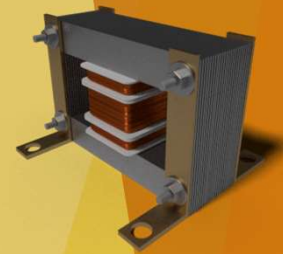
- ▶ When V_1 is applied to the primary and **e.m.f. E_1** is induced known as **primary e.m.f.**

$$E_1 = -N_1 \frac{d\phi}{dt}$$

- ▶ The **e.m.f. E_2** is termed as **secondary e.m.f.**

$$E_2 = -N_2 \frac{d\phi}{dt}$$

$$\frac{E_2}{E_1} = \frac{N_2}{N_1}$$



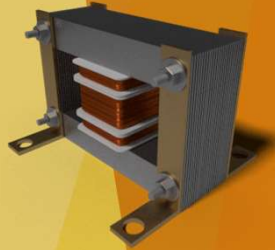
Transformer Type

Step-up transformer

- ▶ If $N_S > N_P$
then $E_S > E_P$
(or $V_S > V_P$)

Step-down transformer

- ▶ If $N_S < N_P$
then $E_S < E_P$
(or $V_S < V_P$)



Ratio

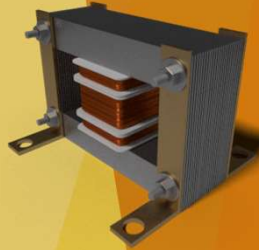
Turns Ratio

a is defined to be the *turns ratio* of the transformer:

$$a = \frac{N_P}{N_S}$$

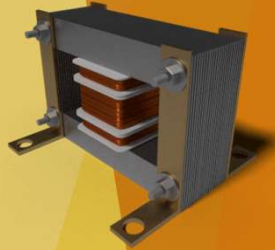
Transformation Ratio

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$



Important Notes!!!

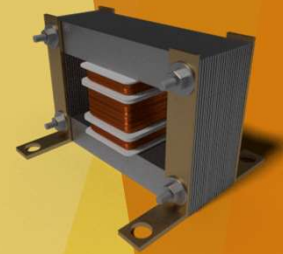
- I. The transformer action is based on the laws of electromagnetic induction.
- II. There is no electrical connection between the primary and secondary.
- III. The a.c. power is transferred from primary to secondary through magnetic flux.
- IV. There is no change in frequency.



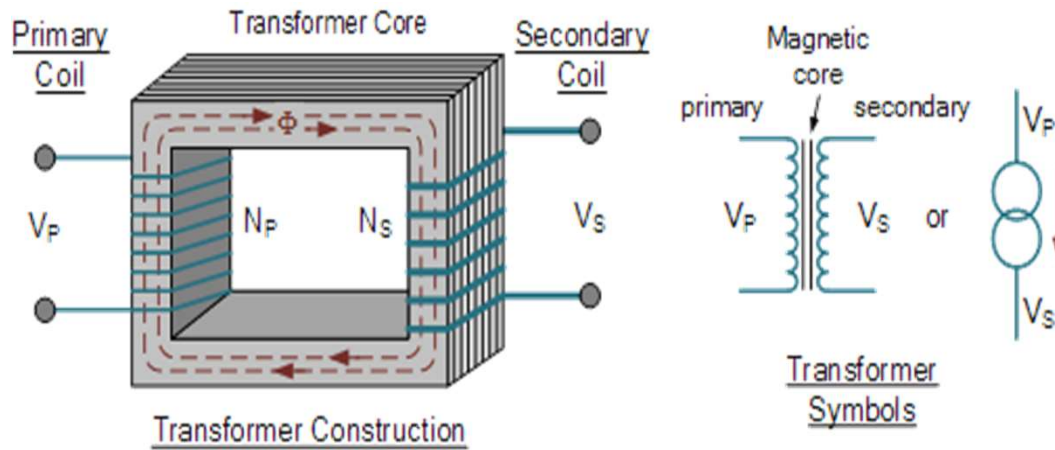
Ideal Transformer

► An ideal transformer is a loss-less device that has

- (i) no winding resistance
- (ii) no leakage flux
- (iii) no iron losses in the core.



Transformer Symbol

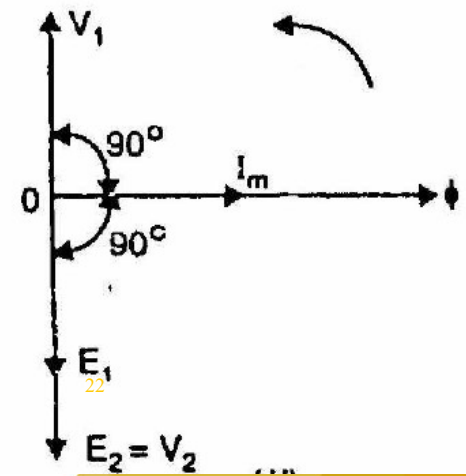
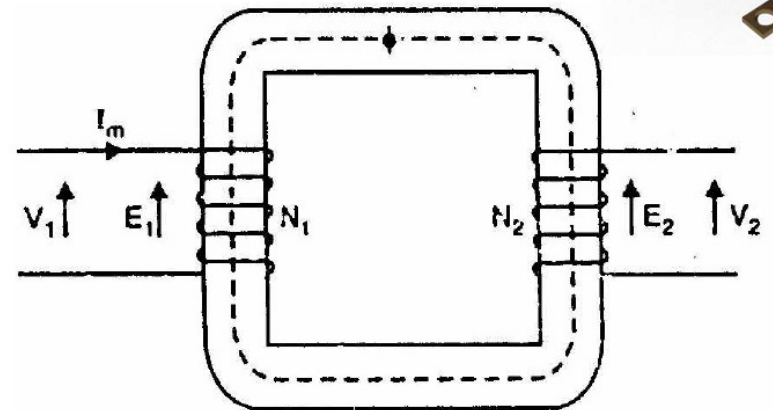


- Where:

- V_P – is the Primary Voltage
- V_S – is the Secondary Voltage
- N_P – is the Number of Primary Windings
- N_S – is the Number of Secondary Windings
- Φ (phi) – is the Flux Linkage

Ideal transformer on - *no load*

- ▶ Alternating voltage V_1 in the primary draws a small magnetizing current I_m .
- ▶ I_m produces an alternating flux ϕ which is proportional to and in phase with it.
- ▶ The alternating flux ϕ links both the windings and induces e.m.f. E_1 in the primary and e.m.f. E_2 in the secondary.
- ▶ The primary e.m.f. E_1 is, at every instant, equal to and in opposition to V_1 (Lenz's law).
- ▶ Both e.m.f.s E_1 and E_2 lag behind flux ϕ by 90° .



Power In Ideal Transformer

Power input at Primary :

$$P_{\text{in}} = V_P I_P \cos \theta_P$$

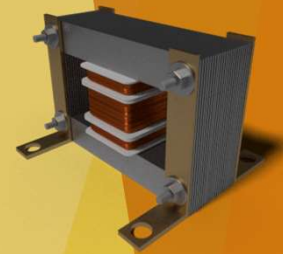
Power output from secondary :

$$P_{\text{out}} = V_S I_S \cos \theta_S$$

Applying the turns-ratio equations gives $V_S = V_P/a$ and $I_S = aI_P$, so

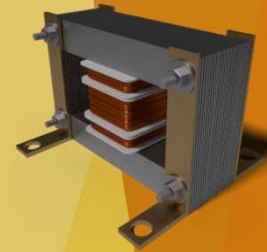
$$P_{\text{out}} = \frac{V_P}{a} (aI_P) \cos \theta$$

$$P_{\text{out}} = V_P I_P \cos \theta = P_{\text{in}}$$



Losses in Practical Transformer

- ▶ *Copper Loss*
- ▶ *Core Loss/ Iron Loss*
- ▶ *Leakage Flux*



Losses in Transformer – *Copper Loss*

- The *resistive heating losses* in the primary and secondary windings of the transformer.
- To reduce the loss *size of the winding* conductor is selected very carefully.
- This loss is proportional to the *square of the current* in the windings.
- To determine the copper losses, *short circuit test* is performed.
- The total copper losses in Transformer are:

$$=I_1^2R_1 + I_2^2R_2$$

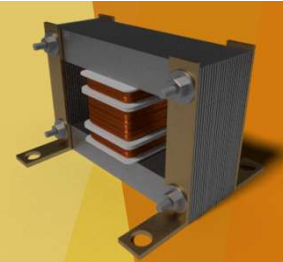


Losses in Transformer – *Iron Loss*

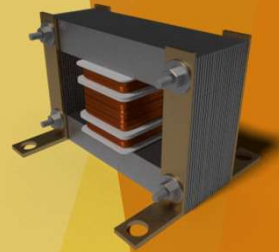
- The power losses that take place in its iron core are known as the '***Iron losses***'.
- These losses occur due to ***alternating flux set up*** in the core.
- To determine the iron losses, open circuit test of transformer is performed.

Iron loss is two types

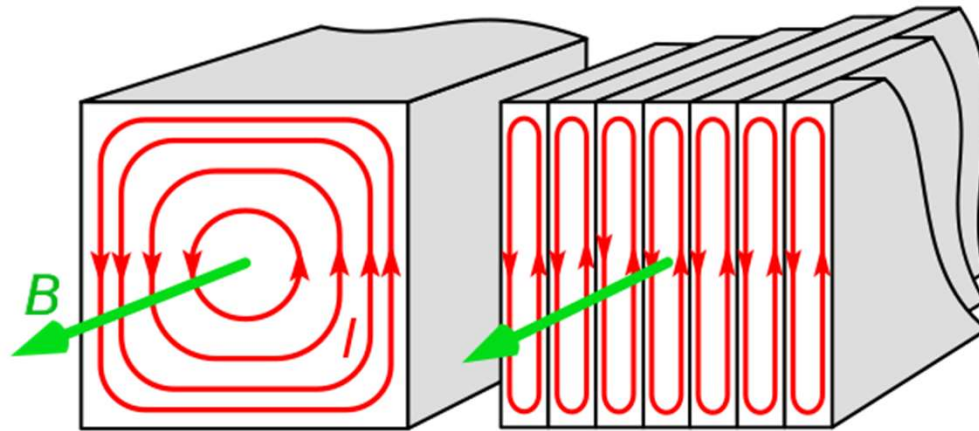
- ***Eddy current losses***
- ***Hysteresis losses***



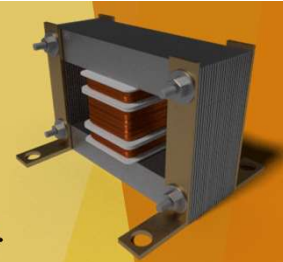
Losses in Transformer – *Eddy Current Loss*



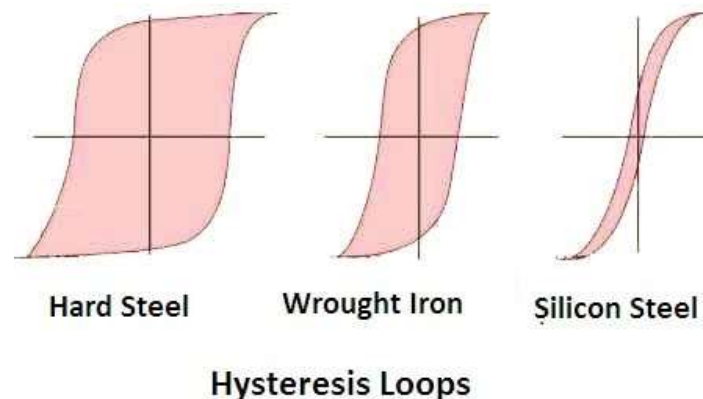
- ▶ Because of time variation of flux in the magnetic material, **current** is induced in the magnetic material. This current is called Eddy current.
- ▶ They are proportional to **the square of the voltage** applied to the transformer.
- ▶ This current is reduced by using laminated (thin sheet) iron structure, with insulation between the laminations.



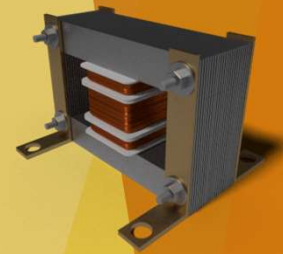
Losses in Transformer – *Hysteresis losses*



- Hysteresis losses are associated with the *rearrangement* of the magnetic domains.
- When a magnetic material is subjected to *alternating flux*, power is required for the continuous reversal of molecular magnets.
- This power is dissipated in the form of heat and is known as '**Hysteresis Loss**'.
- To reduce loss we need to select such type of material which has small hysteresis area.
- The hysteresis loss ($P_h = K_h V f B_m^{1.6}$) is frequency and voltage dependent.

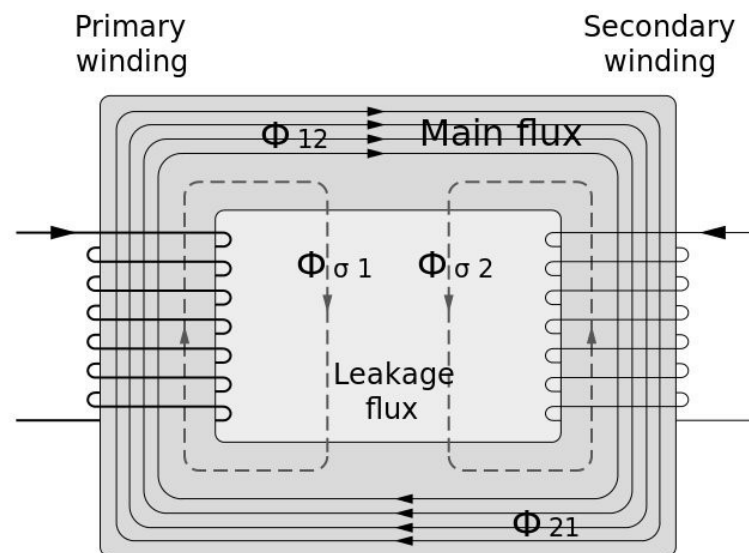


Losses in Transformer – *Leakage flux*

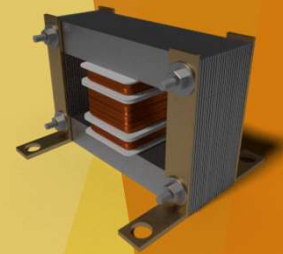


The fluxes which **escape the core** and pass through only one of the transformer windings are leakage fluxes.

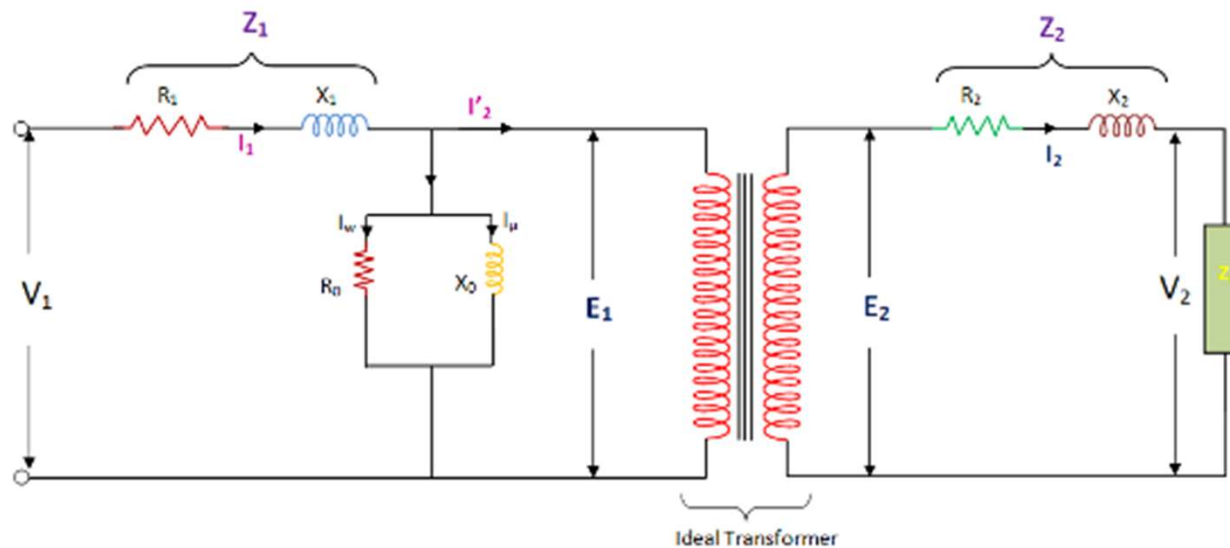
These escaped fluxes produce a *self-inductance* in the primary and secondary coils.



Practical Transformer Equivalent Circuit



- Equivalent circuit diagram of a transformer is basically an equivalent circuit in which the **resistance** and **leakage reactance** of the transformer are imagined to be external to the winding.



Where,
 R_1 = Primary Winding Resistance.
 R_2 = Secondary winding Resistance.
 I_0 = No-load current.
 I_μ = Magnetizing Component,
 I_w = Working Component,

****Equivalent Circuit diagram of Transformer****

Shifting Impedance by *Turns Ratio*

- ▶ The *impedance* of a device is defined:

$$Z_L = \frac{V_L}{I_L}$$

- ▶ Transformer changes the *ratio* between voltage and current and hence the apparent impedance of an element.

- ▶ The impedance of the load is given by :

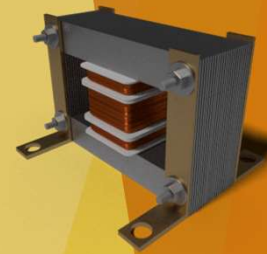
$$Z_L = \frac{V_S}{I_S}$$

- ▶ The apparent impedance of the primary circuit is:

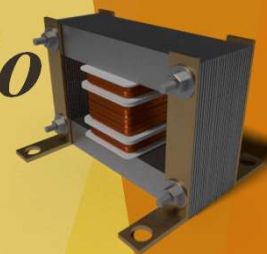
$$Z'_L = \frac{V_P}{I_P}$$

$$Z'_L = \frac{V_P}{I_P} = \frac{aV_S}{I_S/a} = a^2 \frac{V_S}{I_S}$$

$$Z'_L = a^2 Z_L$$



Shifting Impedance by *Transformation Ratio*



$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$

The primary equivalent of the secondary induced voltage is $E_2' = E_2/K = E_1$.

Similarly, primary equivalent of secondary terminal or output voltage is $V_2' = V_2/K$.

Primary equivalent of the secondary current is $I_2' = KI_2$.

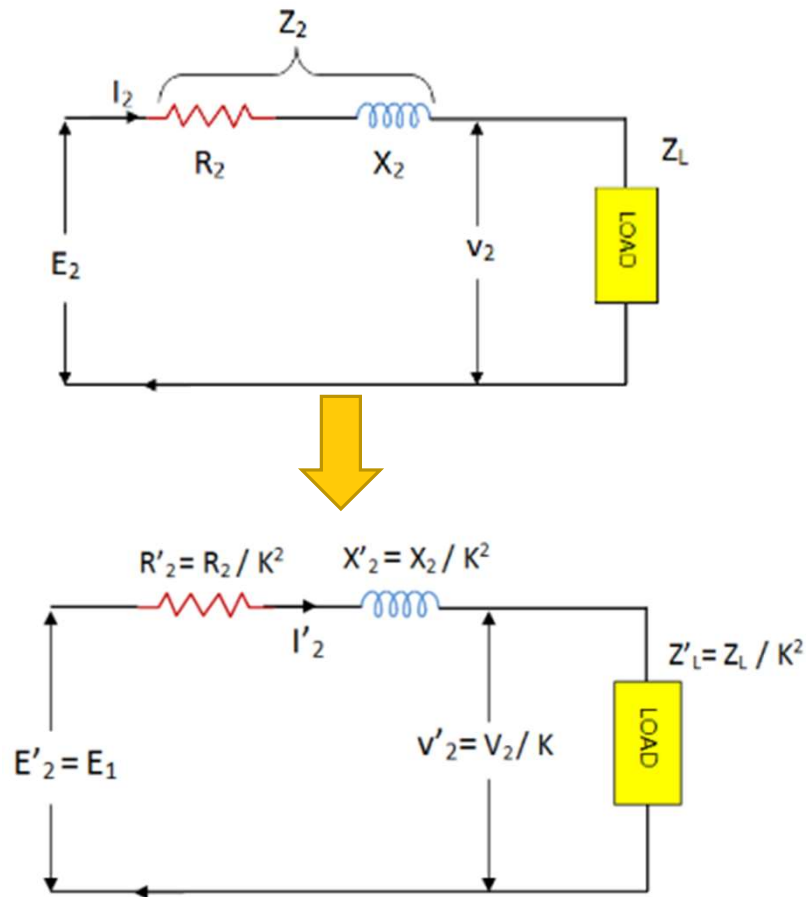
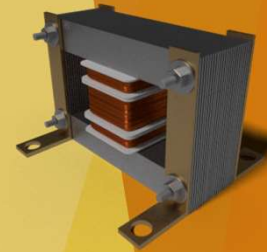
For transferring secondary impedance to primary K^2 is used.

$$R_2' = R_2/K^2$$

$$X_2' = X_2/K^2$$

$$Z_2' = Z_2/K^2$$

Secondary Impedance *Referred to Primary*

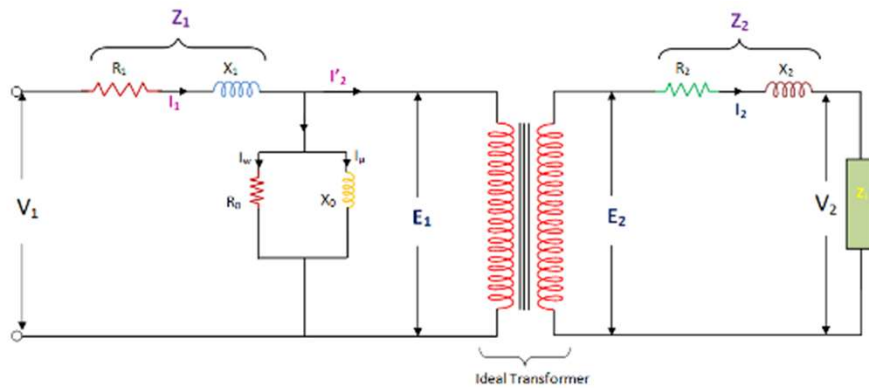
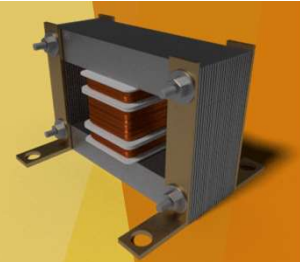


$$R'_2 = R_2 / K^2$$

$$X'_2 = X_2 / K^2$$

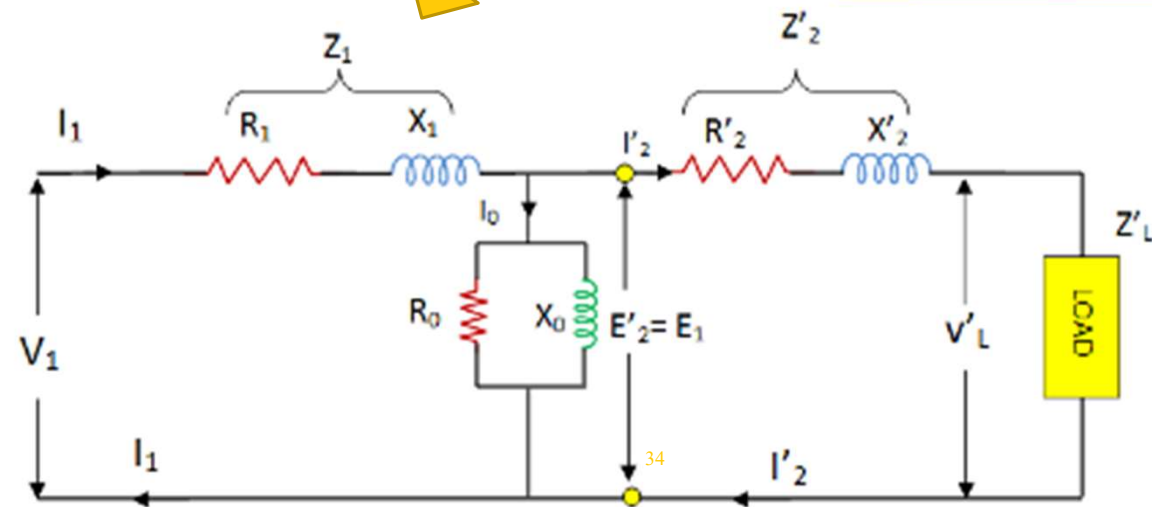
$$Z'_2 = Z_2 / K^2$$

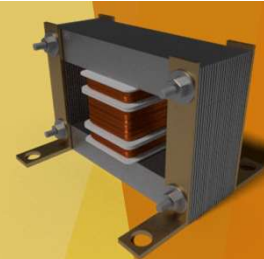
Equivalent Circuit- *Referred to Primary*



****Equivalent Circuit diagram of Transformer****

Equivalent Circuit:

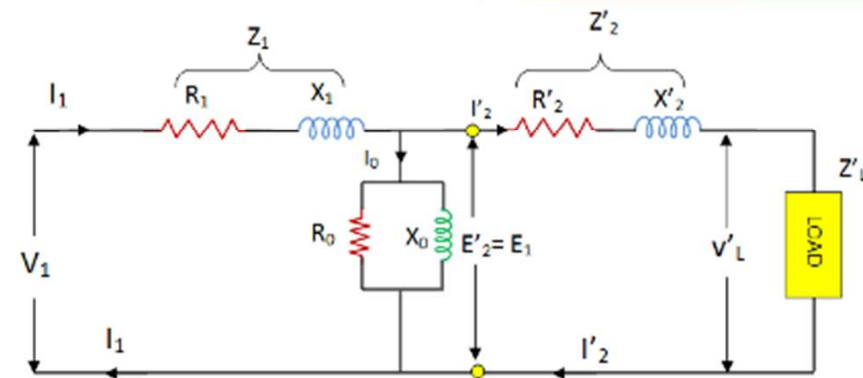




From the equivalent circuit which is shown in fig the total impedance between the input terminal is

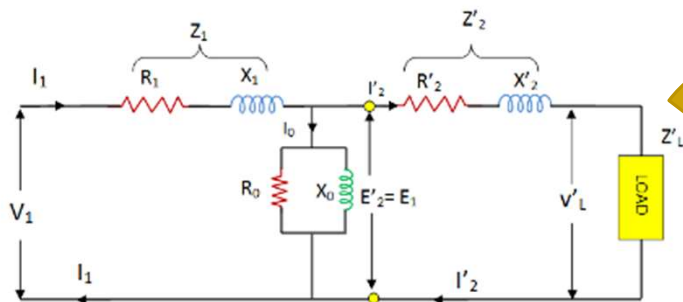
$$Z = Z_1 + Z_m \parallel (Z'_2 + Z'_L) = \left(Z_1 + \frac{Z_m (Z'_m + Z'_L)}{Z_m + (Z'_2 + Z'_L)} \right)$$

$$V_1 = I_1 \left[Z_1 + \frac{Z_m (Z'_m + Z'_L)}{Z_m + (Z'_2 + Z'_L)} \right]$$



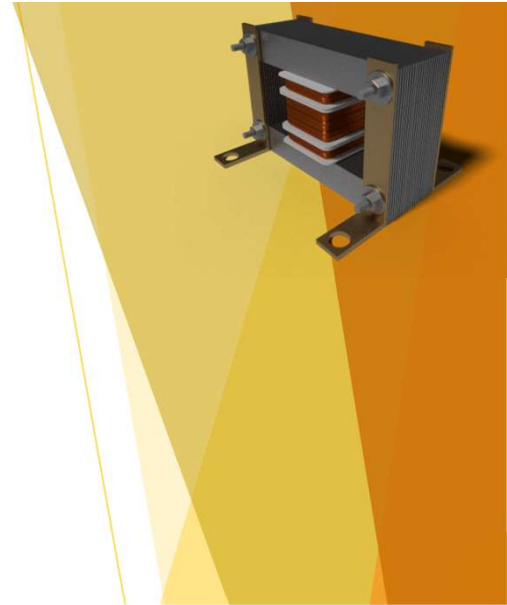
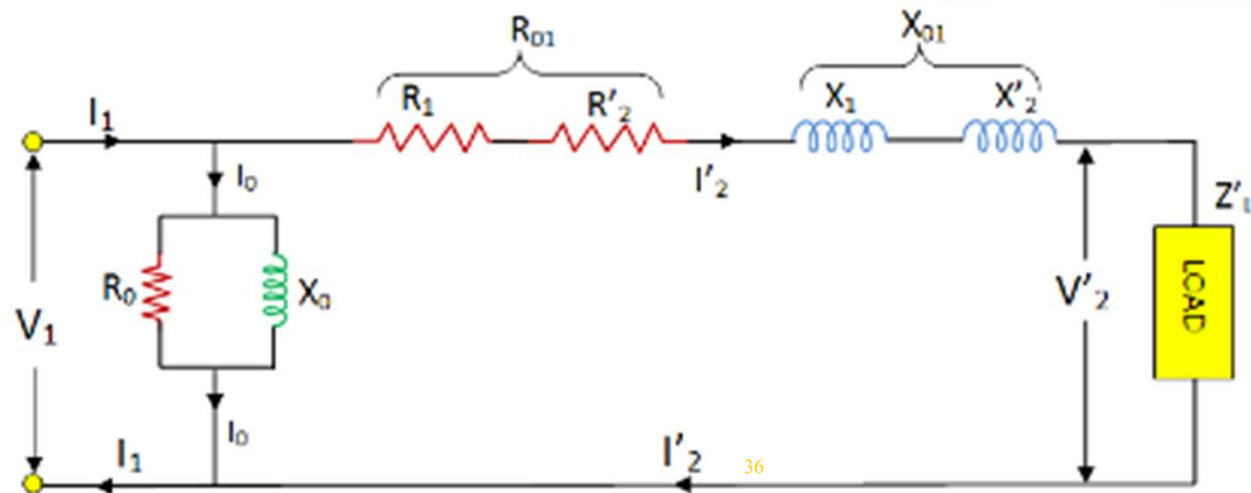
This is so because there are two parallel circuits, one having an impedance of Z_m and the other having Z'_2 and Z'_L in series with each other.

Exact Equivalent circuit



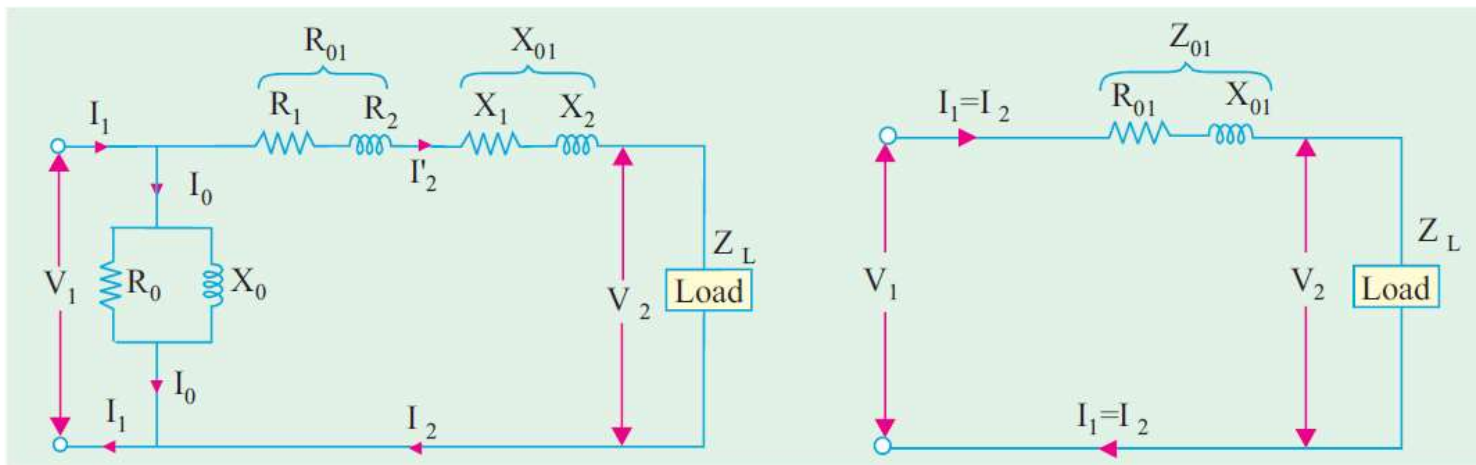
Exact Equivalent circuit:

A simplification can be made by transferring the *exciting circuit* across the terminals.

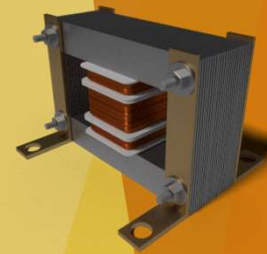


Approximate Equivalent circuit

At last, the circuit is simplified by omitting I_0 altogether as shown in figure below



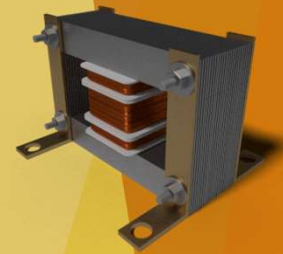
Now, this transformer equivalent circuit is being so easy to analyze.



Voltage Regulation

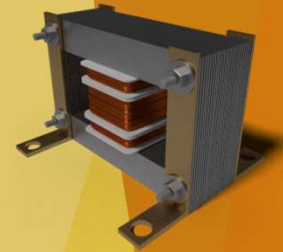
- ▶ The voltage regulation of a transformer is the arithmetic difference between the *no-load* secondary voltage (V_2) and the secondary voltage V_2 *on load* expressed as percentage of no-load voltage.

$$\% \text{age voltage regulation} = \frac{V_2 - V_2}{V_2} \times 100$$



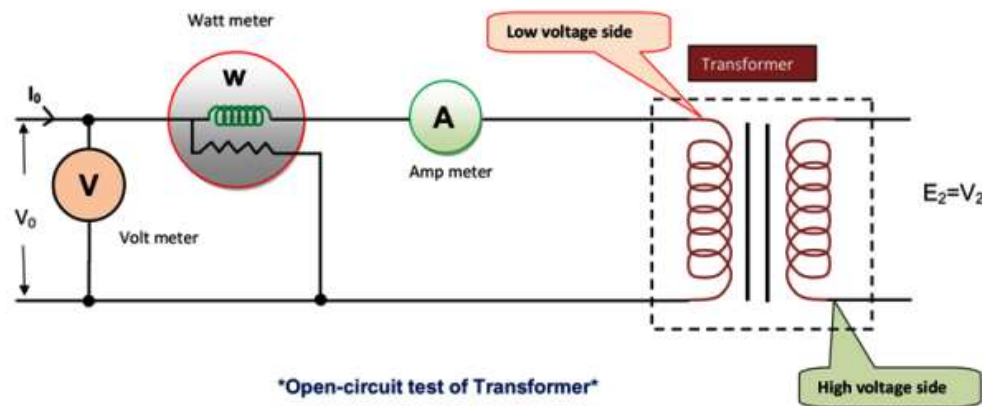
Transformer Tests

- ▶ The performance of a transformer can be calculated on the basis of its equivalent circuit which contains four main parameters.
- ▶ These Parameters can be easily determined by two tests
 - ▶ *Open circuit test or No load test*
 - ▶ *Short circuit test or Impedance test*



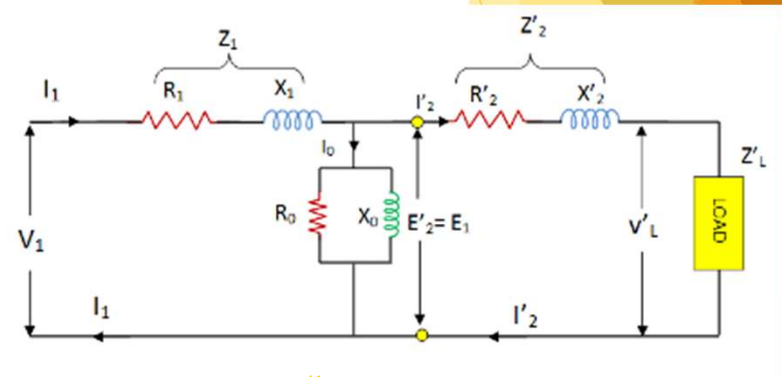
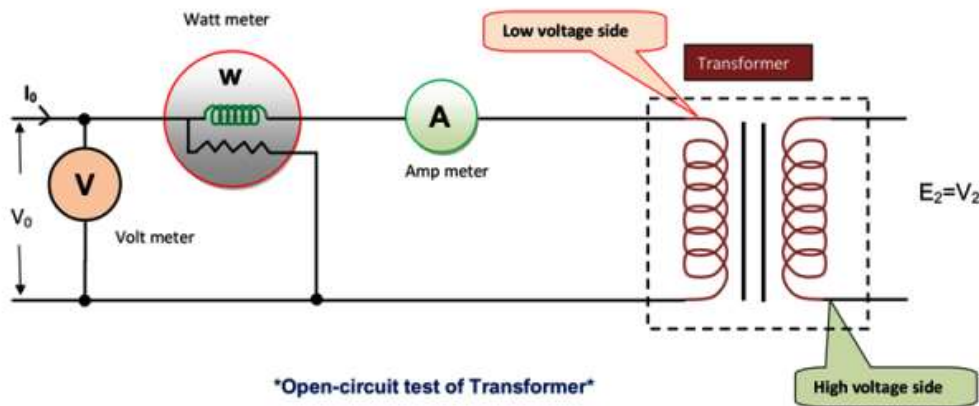
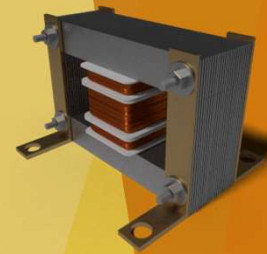
Open Circuit Test

- ▶ Core losses of the transformer can be determined by open circuit test.
- ▶ The high voltage winding is left open.
- ▶ Normal voltage is applied to the low voltage winding.
- ▶ A wattmeter, a voltmeter and an ammeter are connected in the low voltage winding.



Open Circuit Test

- ▶ The Wattmeter measures the iron loss of the transformer.
- ▶ The Ammeter measures the no load current I_0 which is very small (2 to 10 % of rated load current).
- ▶ The voltmeter measures the applied voltage in the low voltage winding.



Open Circuit Test

Let

W is the wattmeter reading

V_1 is the applied voltage

I_0 is the ammeter reading

I_μ is the magnetizing component of no load current

I_w is the core loss component of no load current

Then

$$W = V_1 I_0 \cos \Phi_0$$

$$\therefore \text{No load p.f } \cos \Phi_0 = W / V_1 I_0$$

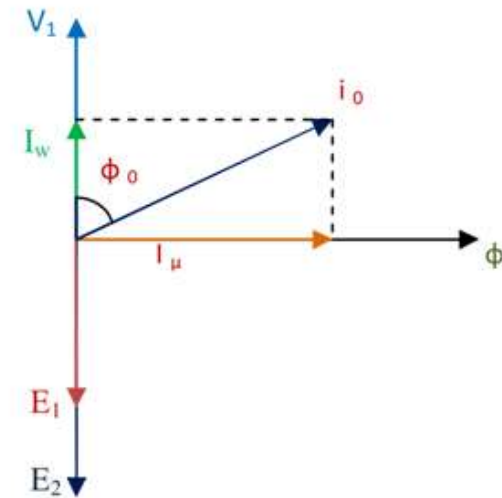
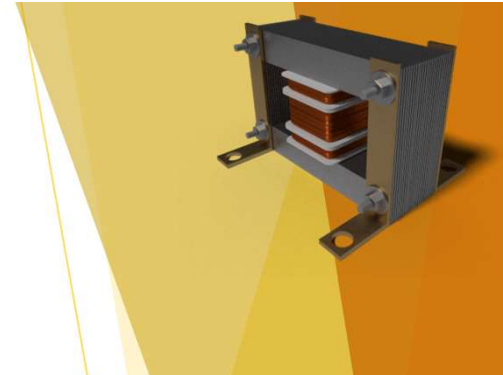
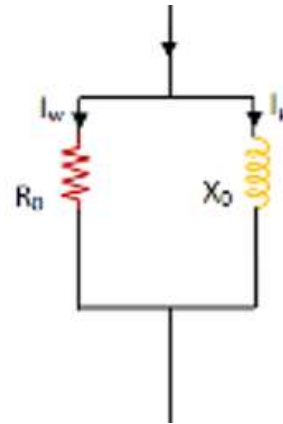


Fig. 1

Open Circuit Test

from the vector diagram of no-load transformer,

$$I_{\mu} = I_0 \sin \Phi_0,$$

$$I_w = I_0 \cos \Phi_0,$$

$$\therefore X_0 = V_1 / I_{\mu} \text{ and}$$

$$R_0 = V_1 / I_w$$

Thus,

$$Z_0 = \sqrt{(R_0^2 + X_0^2)}$$

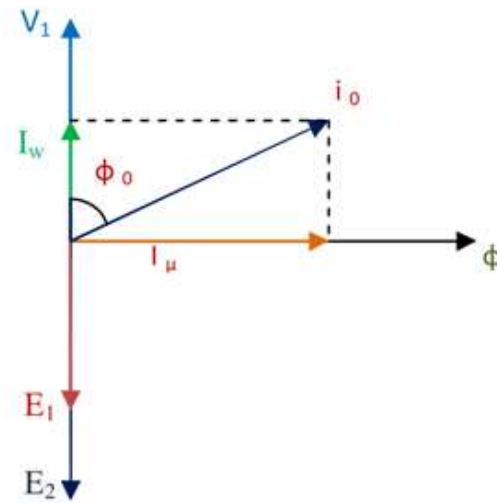
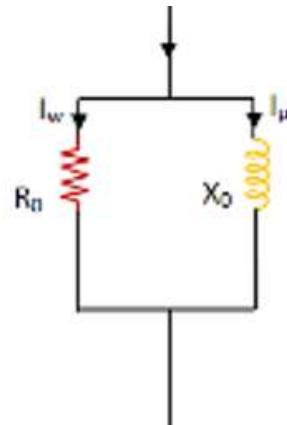
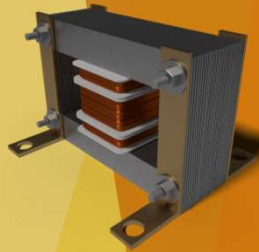
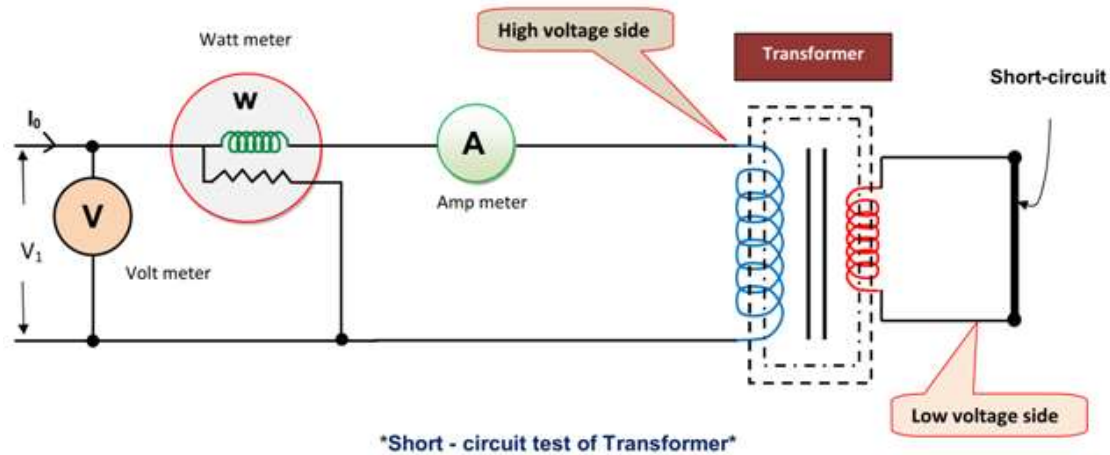


Fig. 1



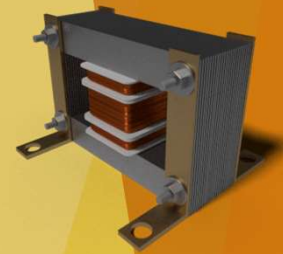
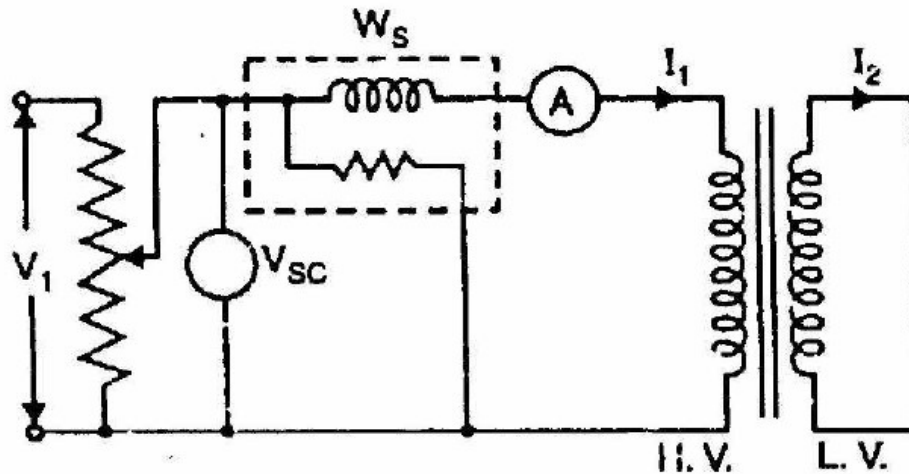
Short Circuit Test

- ▶ This test is conducted to determine R_{01} (or R_{02}), X_{01} (or X_{02}) and full-load copper losses of the transformer.
- ▶ In this test, the low-voltage winding is short-circuited by a thick conductor and normal voltage is applied to the another winding.



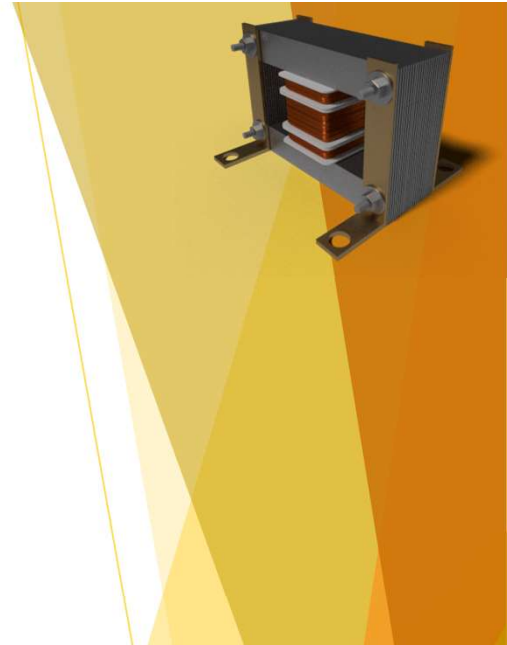
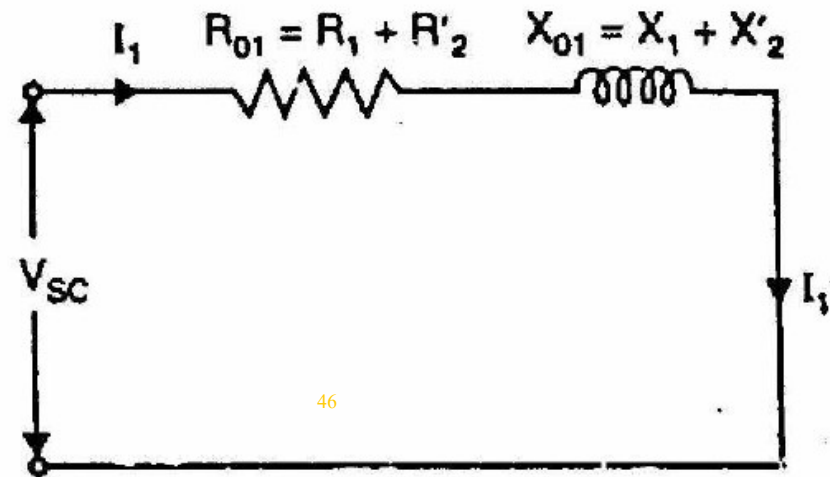
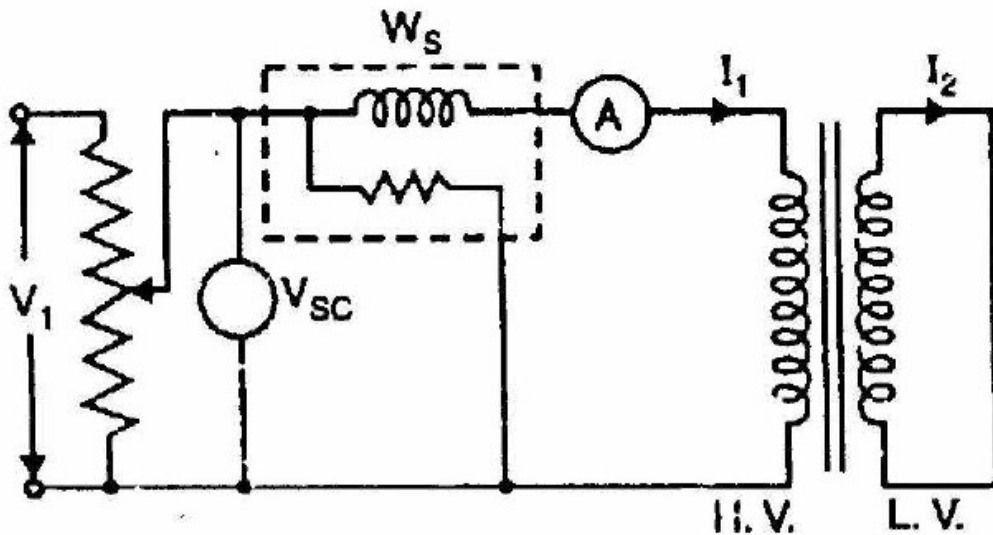
Short Circuit Test

- ▶ The applied voltage is slowly increased until the ammeter gives reading equal to the rated current of the HV side.
- ▶ When the current of HV side reach at rated current of the transformer, wattmeter represents the total full-load Cu-losses of the transformer i.e both primary Cu loss and secondary Cu loss.



Short Circuit Test

- ▶ Full load Cu loss, $P_C = \text{Wattmeter reading} = W_s$
- ▶ Applied voltage = Voltmeter reading = V_{sc}
- ▶ Full Load primary current = Ammeter reading = I_1

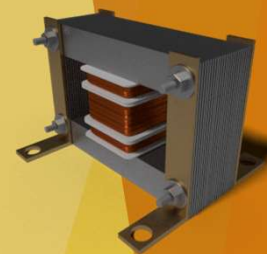
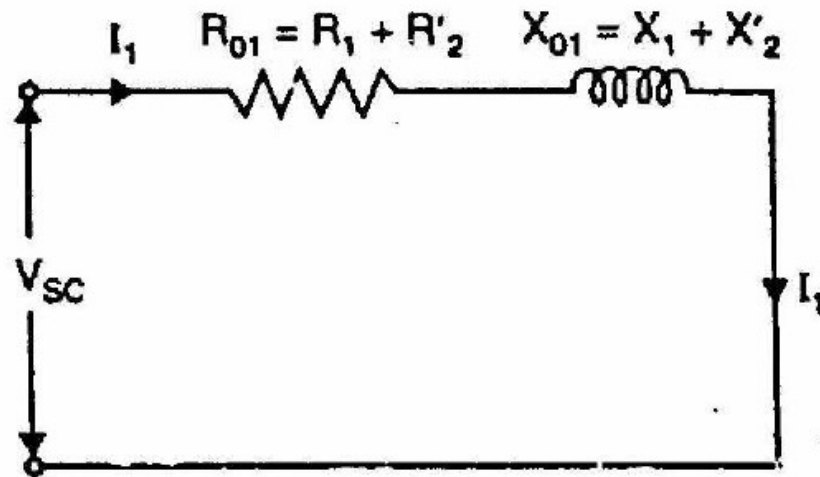


Short Circuit Test

$$P_C = I_1^2 R_1 + I_1^2 R'_2 = I_1^2 R_{01}$$

$$\therefore R_{01} = \frac{P_C}{I_1^2}$$

R_{01} is the total resistance of transformer referred to primary.



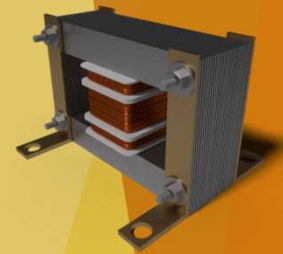
Short Circuit Test

Total impedance referred to primary, $Z_{01} = \frac{V_{SC}}{I_1}$

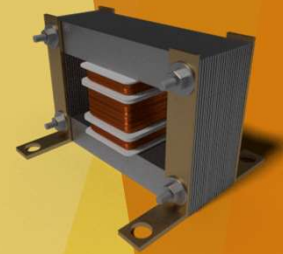
Total leakage reactance referred to primary, $X_{01} = \sqrt{Z_{01}^2 - R_{01}^2}$

Short-circuit p.f, $\cos \phi_2 = \frac{P_C}{V_{SC} I_1}$

Thus short-circuit test gives full-load Cu loss, R_{01} and X_{01}



Advantages of Tests

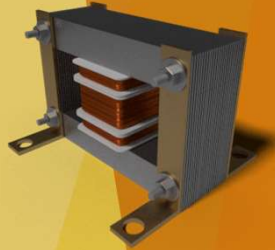


The above two simple transformer tests offer the following advantages:

- (i) The power required to carry out these tests is very small as compared to the full-load output of the transformer.
- (ii) These tests enable us to determine the efficiency of the transformer accurately at any load and p.f. without actually loading the transformer.
- (iii) The short-circuit test enables us to determine R_{01} and X_{01} (or R_{02} and X_{02}). We can thus find the total voltage drop in the transformer as referred to primary or secondary. This permits us to calculate voltage regulation of the transformer.

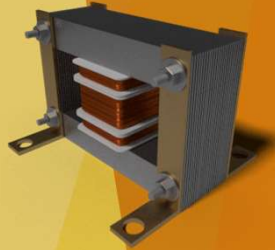
Rating of a Transformer

- ▶ Transformer rating in VA. **But why?**
- ▶ As **copper loss** depends on current and **core loss** depends on voltage.
- ▶ Also we do not know the value or nature of load so we do not know **power factor**.
- ▶ So transformer rated in **VA, KVA, MVA** instead of **KW or MW**.

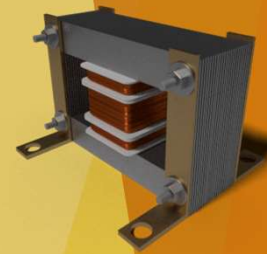
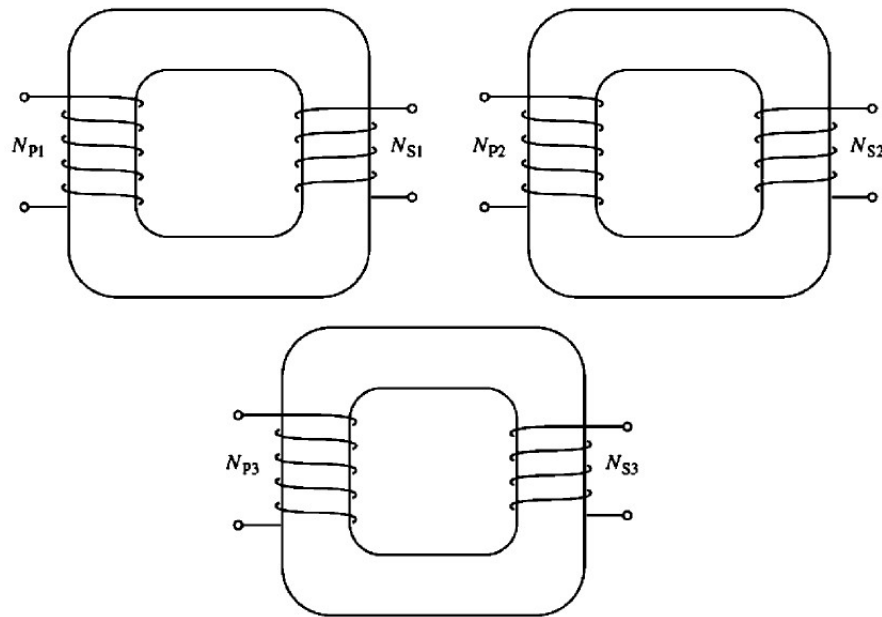


Three Phase Transformer

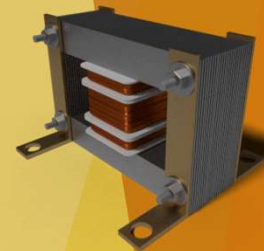
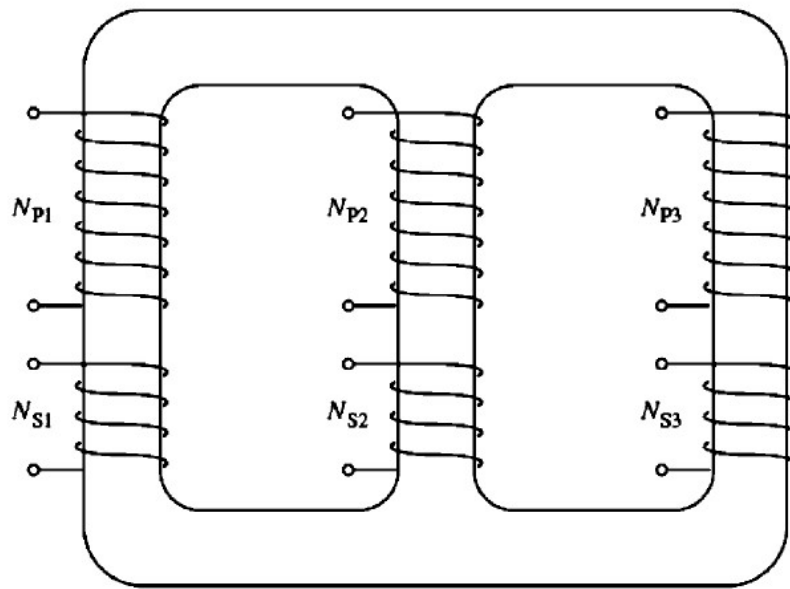
- ▶ Transformers for three-phase circuits can be constructed in one of **two ways**.
- 1. One approach is simply to take three single-phase transformers and connect them in a three-phase bank.
- 2. An alternative approach is to make a three-phase transformer consisting of three sets of windings wrapped on a common core.



Three Phase Transformer

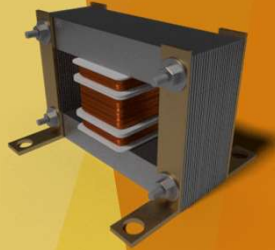


Three Phase Transformer

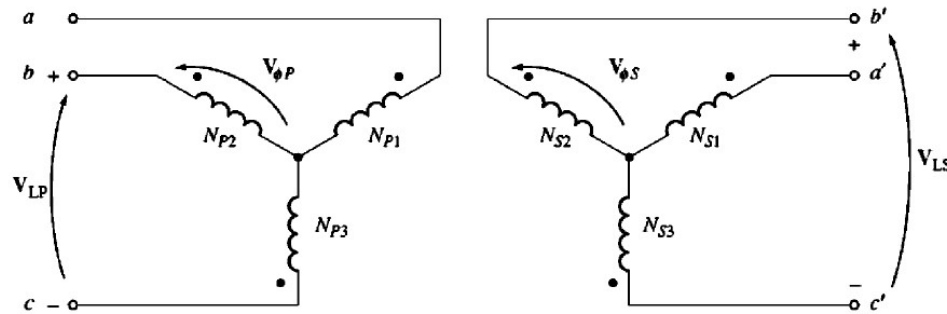


Three Phase Transformer Connection

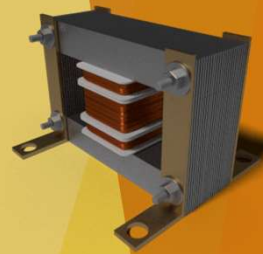
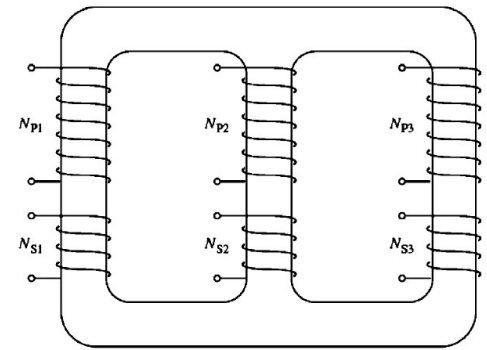
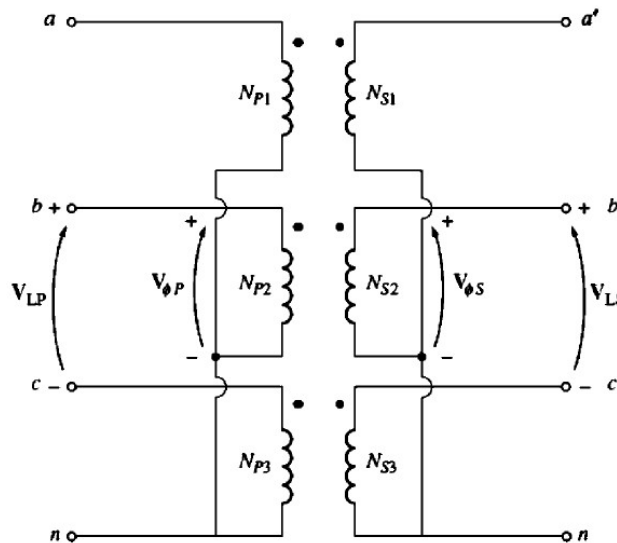
- ▶ The primary or secondary windings of a three phase transformer may be connected in either star (Y) or delta (D) arrangement.
- ▶ The four most common connections are
 - (i) Y-Y
 - (ii) D-D
 - (iii) Y-D and
 - (iv) D-Y.



Y-Y three phase Connection

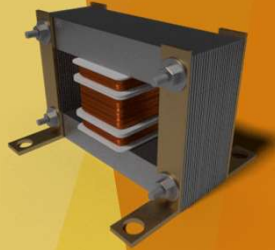


$$\frac{V_{LP}}{V_{LS}} = \frac{\sqrt{3}V_{\phi P}}{\sqrt{3}V_{\phi S}} = a \quad \text{Y-Y}$$



Y-Y connection Problem

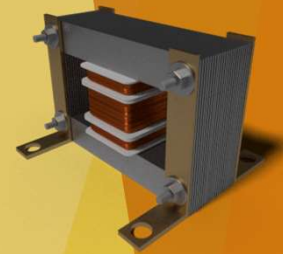
1. If loads on the transformer circuit are unbalanced, then the voltages on the phases of the transformer can become severely unbalanced.
2. Third-harmonic voltages can be large.



Assignment

- ▶ Other three phase connection (Y-D, D-D, D-Y)
 1. construction,
 2. advantages and
 3. disadvantages.

- ▶ Buchholz Relay- function.



Related Videos and references Link

- Faraday' Law of Electromagnetic Induction

<https://www.youtube.com/watch?v=Z0R4TwF2kHo>

<https://www.youtube.com/watch?v=cGCvUPLgygs>

- Electromagnetic Induction and current flow in Transformer

<https://www.youtube.com/watch?v=VucsoEhB0NA>

- Transformer Test

<https://www.youtube.com/watch?v=9hBmgdGjt1Y>

- Transformer Losses

<https://www.youelectricalguide.com/2017/08/power-loss-in-a-transformer.html>

- ▶ Three Phase Transformer Construction

<https://www.youtube.com/watch?v=H2hYUu8IPY0>

